

1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Feature	Ethernet Cable	Fiber Optic Cable
Transmission Medium	Uses copper wires to transmit electrical signals.	Uses glass or plastic fibers to transmit light signals.
Speed & Distance	Suitable for short distances (up to 100 meters) with speeds up to 10 Gbps depending on the cable category.	Supports very high speeds (10 Gbps and above) over long distances, often several kilometers.
Cost	Less expensive and easier to install.	More expensive and requires specialized installation tools and skills.

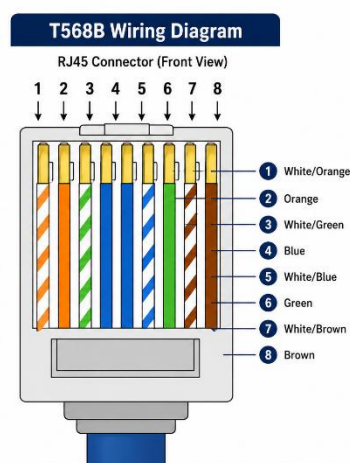
Real-World Examples:

- **Ethernet Cable:** Used to connect a home Wi-Fi router to desktop computers, printers, or gaming consoles.
- **Fiber Optic Cable:** Used for the backbone network in data centers, internet service providers (ISPs), and college campuses where high-speed communication over long distances is required.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

**Straight-Through Ethernet Cable
(T568B Wiring Standard)**

Pin Number	Wire Color
1	White/Orange
2	Orange
3	White/Green
4	Blue
5	White/Blue
6	Green
7	White/Brown
8	Brown



3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

1. Cut the CAT6 cable to the required length using a cable cutter.
2. Strip about 2–3 cm of the outer insulation without damaging the internal wires.
3. Untwist the wire pairs and arrange them in the T568B order:
 - White/Orange
 - Orange
 - White/Green
 - White/Blue
 - Green
 - White/Brown
 - Brown
4. Straighten the wires and trim them evenly.
5. Insert the wires carefully into the RJ45 connector until each wire reaches the end of the connector.
6. Place the RJ45 connector into the RJ45 crimping tool.
7. Squeeze the crimper firmly to secure the connector onto the cable.
8. Test the finished cable using a network cable tester to ensure all eight wires are connected correctly.

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4

sentences.

Hint: Consider speed, distance, and budget for your answer.

For a gaming café with 10 computers, I would use **CAT6 Ethernet cables** and an **RJ45 crimping tool**. Ethernet cables provide excellent speed, low latency, and are cost-effective for distances within a single room or building. Fiber optic cables are designed for long-distance, high-capacity connections and would increase the installation cost unnecessarily. Therefore, CAT6 Ethernet cables with RJ45 connectors are the most practical and economical choice for a LAN gaming setup.